C9200_9300_9400_95... × C9200_9300_9400_95... ×

```
00000000  A3 88 06 41 54 0B D6 55 75 1B 69 AB EC E7 30 EA  ╜ê.AT.╓Uu.i½∞τ0Ω
00000010  8E 33 3A 8D E8 F8 C1 DC 45 29 72 9B EA 56 6E 8D  Ä3:iφ°└▄E)r¢ΩVni
00000020  F2 DB 56 0B E9 79 C0 5A 0A 74 8F 1A B4 B3 37 0A  ≥╠V.⊖y└Z.tÅ.┤│7.
00000030  88 06 87 C8 00 00 00 00 DE AD BE EF DE AD BE EF  ê.ç╚....▐¡╝∩▐¡╝∩
00000040  00 00 04 0C 00 00 75 30 00 00 00 00 02 7D 88 F9  ......u0.....}ê·
00000050  31 1E F8 F9 00 00 00 00 00 00 00 00 00 00 00 00  1.°·............
00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03  ................
00000090  00 00 00 09 00 00 00 1D 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
000000A0  50 41 43 4B 41 47 45 5F 43 4F 4D 50 41 54 49 42  PACKAGE_COMPATIB
000000B0  49 4C 49 54 59 00 00 00 00 00 00 09 00 00 00 0B  ILITY...........
000000C0  46 52 55 5F 52 50 5F 54 59 50 45 00 00 00 00 09  FRU_RP_TYPE.....
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 0A  EYSTRING........
00000160  45 6E 43 72 59 70 54 69 4F 6E 00 00 00 00 00 09  EnCrYpTiOn......
00000170  00 00 00 0B 43 57 5F 42 45 47 49 4E 3D 24 24 00  ....CW_BEGIN=$$.
00000180  00 00 00 09 00 00 00 11 43 57 5F 46 41 4D 49 4C  ........CW_FAMIL
00000190  59 3D 24 63 61 74 39 6B 24 00 00 00 00 00 00 09  Y=$cat9k$.......
000001A0  00 00 00 27 43 57 5F 49 4D 41 47 45 3D 24 63 61  ...'CW_IMAGE=$ca
000001B0  74 39 6B 5F 69 6F 73 78 65 2E 31 36 2E 31 32 2E  t9k_iosxe.16.12.
000001C0  30 37 2E 53 50 41 2E 62 69 6E 24 00 00 00 00 00  07.SPA.bin$.....
000001D0  00 00 00 15 43 57 5F 56 45 52 53 49 4F 4E 3D 24  ....CW_VERSION=$
000001E0  31 36 2E 31 32 2E 30 37 24 00 00 00 00 00 00 09  16.12.07$.......
000001F0  00 00 00 37 43 57 5F 46 55 4C 4C 5F 56 45 52 53  ...7CW_FULL_VERS
00000200  49 4F 4E 3D 24 31 36 2E 31 32 2E 30 37 2E 30 2E  ION=$16.12.07.0.
00000210  36 35 36 35 2E 31 36 34 33 38 32 39 39 39 36 2E  6565.1643829996.
00000220  2E 47 69 62 72 61 6C 74 61 72 24 00 00 00 00 09  .Gibraltar$.....
00000230  00 00 00 34 43 57 5F 44 45 53 43 52 49 50 54 49  ...4CW_DESCRIPTI
```

C9200_9300_9400_95... × C9200_9300_9400_95... ×

```
00000240  4F 4E 3D 24 43 69 73 63 6F 20 49 4F 53 20 53 6F  ON=$Cisco IOS So
00000250  66 74 77 61 72 65 2C 20 49 4F 53 2D 58 45 20 53  ftware, IOS-XE S
00000260  6F 66 74 77 61 72 65 24 00 00 00 09 00 00 00 09  oftware$........
00000270  43 57 5F 45 4E 44 3D 24 24 00 00 00 00 00 00 0C  CW_END=$$.......
00000280  00 00 01 88 01 00 02 01 01 02 00 04 00 00 01 7C  ...ê...........|
00000290  03 00 61 04 00 28 43 4E 3D 43 69 73 63 6F 53 79  ..a..(CN=CiscoSy
000002A0  73 74 65 6D 73 3B 4F 55 3D 49 4F 53 2D 58 45 3B  stems;OU=IOS-XE;
000002B0  4F 3D 43 69 73 63 6F 53 79 73 74 65 6D 73 05 00  O=CiscoSystems..
000002C0  08 36 31 46 41 45 35 39 42 06 00 28 43 4E 3D 43  .61FAE59B..(CN=C
000002D0  69 73 63 6F 53 79 73 74 65 6D 73 3B 4F 55 3D 49  iscoSystems;OU=I
000002E0  4F 53 2D 58 45 3B 4F 3D 43 69 73 63 6F 53 79 73  OS-XE;O=CiscoSys
000002F0  74 65 6D 73 07 00 01 00 08 00 01 01 09 00 01 00  tems............
00000300  0A 00 01 01 0B 01 00 6A 47 40 19 8C E2 A1 A5 AA  .......jG@.îΓíÑ¬
00000310  1D 34 FE 1C DB CF BF 5C 0A 14 56 92 A6 03 5B B1  .4▪.╙╧┐\..VÆª.[▒
00000320  0E 86 9A 8F 3D EB B7 92 C0 B9 85 95 C4 85 2A 95  .åÜÅ=δ╗Æ╚╢àò╣à*ò
00000330  22 18 E6 32 E8 2A 29 16 05 92 98 24 C6 76 CC 8C  ".µ2Φ*)..Æÿÿÿÿÿÿÿ
00000340  60 D3 BB 94 AB D9 6E 63 83 61 4D AE 37 20 41 D6  `╙╗ö½╝ncâaM«7 A╓
00000350  B9 76 2A 57 D0 E5 12 25 83 F4 3A 53 B3 5B 6F 53  ╣v*W╨σ.%â⌠:S│[oS
00000360  68 F4 E7 F3 BF 77 28 99 61 36 24 B3 0C C6 A4 54  h⌠τ≤┐w(Öa6$│.╞¤T
00000370  0F 44 8A 8A 93 36 CD 07 C9 25 14 79 8D 5B 53 96  .DèèôÖ6═.╔%.yì⌡Sû
00000380  31 F7 28 C2 EE 95 51 34 00 09 E3 56 8E 4E CA D1  1≈(┬εòQ4..πVÄN╩╤
00000390  3E CB EB 97 B5 18 CC 6B 7E 7F AB C2 EF E8 92 6D  >╦δù╡.╠k~⌂½┬∩Φæm
000003A0  63 04 5F 4B 47 52 22 BF 87 22 5F 07 0A CA FB 24  c._KGR"┐ç"_..╩√$
000003B0  B3 53 41 14 96 BF 3D AA FE 5F 2F 1F 4C E0 E3 67  │SA.û┐=¬▪_/.Lαπg
000003C0  3F 70 9B F6 96 AF 08 E4 3B 8B 5A 34 F1 93 F6 C3  ?p¢÷û».Σ;ïZ4±ô÷╠
000003D0  C5 FB 12 DE 5A CF 3F 0A AB 0E 2D 34 62 A0 64 98  ┼√.▐Z╧?.½.-4bádÿ
000003E0  C2 42 0C 77 AD 37 01 AA FD AA 8B 66 36 7B 8D 0D  ┬B.w¡7.¬²¬ïf6{ì.
000003F0  EB DF 75 34 DA 78 49 F0 B5 EC DB E2 35 A6 AB E3  δ▀u4┌xI≡╡∞▐ⁿ5ª½π
00000400  97 14 EB 3F 0C 35 AF 0C 00 01 41 EB 5C F0 F1 19  ù.δ?.5».¬.Aδ\≡±.
00000410  61 B6 86 96 3A D7 8A 4F 0F EF E6 19 F8 21 DC 73  a╢åû:╫èO.∩µ.°!▄s
00000420  CC CC EA 46 E2 09 7B DC 96 29 6C 43 2B 88 3D A3  ╠╠╩FΓ.{▄û)lC+ê=ú
00000430  83 5D C6 2C 26 90 0C AA 10 2E 69 8B 75 2D 96 92  â]╞,&É.¬..ïïu-ûÆ
00000440  00 00 00 00 DE AD BE EF DE AD BE EF 00 00 04 04  ....▐¡╝∩▐¡╝∩....
00000450  00 00 75 31 00 00 00 00 00 4E 96 14 02 7D 84 ED  ..u1.....Nû..}äφ
00000460  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000470  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

C9200_9300_9400_95… × C9200_9300_9400_95… ×

```
00000480  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000490  00 00 00 00 00 00 00 00 00 00 00 03 00 00 00 09  ................
000004A0  00 00 00 1D 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000004B0  41 47 45 5F 43 4F 4D 50 41 54 49 42 49 4C 49 54  AGE_COMPATIBILIT
000004C0  59 00 00 00 00 00 00 09 00 00 00 0B 46 52 55 5F  Y...........FRU_
000004D0  52 50 5F 54 59 50 45 00 00 00 00 09 00 00 00 18  RP_TYPE.........
000004E0  4B 45 59 5F 54 4C 56 5F 50 41 43 4B 41 47 45 5F  KEY_TLV_PACKAGE_
000004F0  42 4F 4F 54 41 52 43 48 00 00 00 09 00 00 00 0E  BOOTARCH........
00000500  41 52 43 48 5F 69 36 38 36 5F 54 59 50 45 00 00  ARCH_i686_TYPE..
00000510  00 00 00 09 00 00 00 14 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
00000520  42 4F 41 52 44 5F 43 4F 4D 50 41 54 00 00 00 09  BOARD_COMPAT....
00000530  00 00 00 10 42 4F 41 52 44 5F 63 61 74 39 6B 5F  ....BOARD_cat9k_
00000540  54 59 50 45 00 00 00 09 00 00 00 18 4B 45 59 5F  TYPE........KEY_
00000550  54 4C 56 5F 43 52 59 50 54 4F 5F 4B 45 59 53 54  TLV_CRYPTO_KEYST
00000560  52 49 4E 47 00 00 00 09 00 00 00 04 6E 6F 6E 65  RING........none
00000570  00 00 00 09 00 00 00 0B 43 57 5F 42 45 47 49 4E  ........CW_BEGIN
00000580  3D 24 24 00 00 00 00 09 00 00 00 11 43 57 5F 46  =$$.........CW_F
00000590  41 4D 49 4C 59 3D 24 63 61 74 39 6B 24 00 00 00  AMILY=$cat9k$...
000005A0  00 00 00 09 00 00 00 28 43 57 5F 49 4D 41 47 45  .......(CW_IMAGE
000005B0  3D 24 63 61 74 39 6B 2D 72 70 62 6F 6F 74 2E 31  =$cat9k-rpboot.1
000005C0  36 2E 31 32 2E 30 37 2E 53 50 41 2E 70 6B 67 24  6.12.07.SPA.pkg$
000005D0  00 00 00 09 00 00 00 15 43 57 5F 56 45 52 53 49  ........CW_VERSI
000005E0  4F 4E 3D 24 31 36 2E 31 32 2E 30 37 24 00 00 00  ON=$16.12.07$...
000005F0  00 00 00 09 00 00 00 37 43 57 5F 46 55 4C 4C 5F  .......7CW_FULL_
00000600  56 45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30  VERSION=$16.12.0
00000610  37 2E 30 2E 36 35 36 35 2E 31 36 34 33 38 32 39  7.0.6565.1643829
00000620  39 39 36 2E 2E 47 69 62 72 61 6C 74 61 72 24 00  996..Gibraltar$.
00000630  00 00 00 09 00 00 00 34 43 57 5F 44 45 53 43 52  .......4CW_DESCR
00000640  49 50 54 49 4F 4E 3D 24 43 69 73 63 6F 20 49 4F  IPTION=$Cisco IO
00000650  53 20 53 6F 66 74 77 61 72 65 2C 20 49 4F 53 2D  S Software, IOS-
00000660  58 45 20 53 6F 66 74 77 61 72 65 24 00 00 00 09  XE Software$....
00000670  00 00 00 09 43 57 5F 45 4E 44 3D 24 24 00 00 00  ....CW_END=$$...
00000680  00 00 00 0C 00 00 01 88 01 00 02 01 01 02 00 04  .......ê........
00000690  00 00 01 7C 03 00 61 04 00 28 43 4E 3D 43 69 73  ...|..a..(CN=Cis
000006A0  63 6F 53 79 73 74 65 6D 73 3B 4F 55 3D 49 4F 53  coSystems;OU=IOS
000006B0  2D 58 45 3B 4F 3D 43 69 73 63 6F 53 79 73 74 65  -XE;O=CiscoSyste
```

C9200_9300_9400_95… × C9200_9300_9400_95… ×

```
000006C0  6D 73 05 00 08 36 31 46 41 45 34 46 42 06 00 28  ms...61FAE4FB..(
000006D0  43 4E 3D 43 69 73 63 6F 53 79 73 74 65 6D 73 3B  CN=CiscoSystems;
000006E0  4F 55 3D 49 4F 53 2D 58 45 3B 4F 3D 43 69 73 63  OU=IOS-XE;O=Cisc
000006F0  6F 53 79 73 74 65 6D 73 07 00 01 00 08 00 01 01  oSystems........
00000700  09 00 01 00 0A 00 01 01 0B 01 00 B8 3A FE D4 52  ...........╕:■╙R
00000710  30 29 CD 27 13 BF CD AB 04 CA 82 20 2C A2 DE 2F  0)═'.┐═½.╝é ,ó▌/
00000720  6B B2 49 2A 7A 70 4E 94 38 A4 93 58 45 ED BD D8  k▒I*zpNö8ñôXEφ╝╪
00000730  0A 19 36 8A 03 3F 7A D1 BA 04 2A 8F AB 31 53 5D  ..6è.?z╤╢.*Å½1S]
00000740  92 5D 0D 03 C2 4F 36 99 3C D8 4B CA 03 08 B6 67  Æ]..┬O6Ö<╪K╩..╢g
00000750  58 BC 55 BA A0 1E 28 2C BF 9F 0A C8 D2 AE 71 94  X╝U╢á.(,┐ƒ.╚╥®qö
00000760  23 80 C5 F5 79 61 85 80 0B 25 64 63 D0 59 25 A1  #Ç╕┤yaàÇ.%dc╨Y%í
00000770  55 18 40 B4 E3 43 3C 17 5A 8D 02 2E E5 71 5D 24  U.@├πC<.Zì..σq]$
00000780  31 74 51 55 D0 31 9F 24 F9 56 65 DD B4 07 04 61  1tQU╨1ƒ$·Ve▌┤..a
00000790  49 2D A4 20 9B FD 90 D4 F0 64 38 60 70 C4 8A 42  I-ñ ¢²É╚≡d8`p─èB
000007A0  D4 F3 9C 4C 71 29 CF 73 19 D5 88 DA 6E 6C EB 87  ╘≤£Lq)╧s.╒ê┌nlδç
000007B0  59 F0 18 27 3C 4F C1 31 D3 50 B2 81 A1 C3 46 76  Y≡.'<O┴1╙P▒üí├Fv
000007C0  91 B2 D7 C7 6B 6A 48 43 BF E2 0A 3B B6 CA 1D DA  æ▒╫╟kjHC┐Γ.;╢╩.┌
000007D0  CB 41 91 E0 8C 04 5D CD 3E 66 A0 E7 67 09 10 6F  ╦Aæαî.]═>fá╕g..o
000007E0  2C 6C C8 9A 79 68 3D DB A4 35 20 6E 55 4B 2C 00  ,l╚Üyh=█ñ5 nUK,.
000007F0  DF 01 83 6E ED 03 AB 0E 7D 9A 4B 0B 58 9F 64 F9  ▀.ânφ.½.}ÜK.Xƒd·
00000800  C3 AA 86 1F D8 87 07 EF F5 26 1F 0C 00 01 41 EB  ├¬å.╪ç.∩⌡&....Aδ
00000810  EA 05 00 C0 07 8C C8 8E D8 8E C0 8E D0 31 E4 FB  Ω..└.î╚Ä╪Ä└Ä╨1Σ√
00000820  FC BE 2D 00 AC 20 C0 74 09 B4 0E BB 07 00 CD 10  ⁿ╝-.¼ └t.┤.╗..═.
00000830  EB F2 31 C0 CD 16 CD 19 EA F0 FF 00 F0 55 73 65  δ≥1└═.═.Ω≡ .≡Use
00000840  20 61 20 62 6F 6F 74 20 6C 6F 61 64 65 72 2E 0D   a boot loader..
00000850  0A 0A 52 65 6D 6F 76 65 20 64 69 73 6B 20 61 6E  ..Remove disk an
00000860  64 20 70 72 65 73 73 20 61 6E 79 20 6B 65 79 20  d press any key
00000870  74 6F 20 72 65 62 6F 6F 74 2E 2E 2E 0D 0A 00 00  to reboot.......
00000880  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000890  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
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00000900   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000910   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000920   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000930   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000940   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000950   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000960   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000970   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000980   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
00000990   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009A0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009B0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009C0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009D0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009E0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
000009F0   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF   ...............
00000A00   FF 1E 01 00 41 E5 04 00 00 00 FF FF 00 00 55 AA    ...Aσ....  ..U¬
00000A10   EB 66 48 64 72 53 0D 02 00 00 00 00 00 10 00 30   δfHdrS.........0
00000A20   00 01 00 80 00 00 10 00 00 00 00 00 00 00 00 00   ...Ç...........
00000A30   00 00 00 00 80 61 00 00 00 00 00 00 FF FF FF 7F   ....Ça......   ⌂
00000A40   00 00 20 00 01 15 13 00 FF 07 00 00 00 00 00 00   .. ..... .......
00000A50   00 00 00 00 00 00 00 00 6B 02 00 00 34 E8 4D 00   ........k...4ΦM.
00000A60   00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00   ................
00000A70   00 C0 2D 01 00 00 00 00 8C D8 8E C0 FC 8C D2 39   .└-.....î╬Ä└ⁿî╥9
00000A80   C2 89 E2 74 16 BA 80 5F F6 06 11 02 80 74 04 8B   ┬ëΓt.│Ç_÷...Çt.ï
00000A90   16 24 02 81 C2 00 04 73 02 31 D2 83 E2 FC 75 03   .$.ü┬..s.1╥âΓⁿu.
00000AA0   BA FC FF 8E D0 66 0F B7 E2 FB 1E 68 9F 02 CB 66   │ⁿ Ä╨f.╗Γ√.hƒ.╦f
00000AB0   81 3E 58 3C 55 AA 5A 5A 75 17 BF 60 3C B9 83 5F   ü>X<U¬ZZu.┐`<╣â_
00000AC0   66 31 C0 29 F9 C1 E9 02 F3 66 AB 66 E8 32 29 00   f1└)·┴Θ.≤f½fΦ2).
00000AD0   00 66 B8 D2 03 00 00 66 E8 E9 00 00 00 F4 EB FD   .f╕╥...fΦΘ...⌠δ²
00000AE0   38 06 FB 02 74 05 A2 FB 02 EB 00 66 9C 0F A0 0F   8.√.t.ó√.δ.f£.á.
00000AF0   A8 66 60 83 EC 2C 89 D6 89 E7 B9 0B 00 F3 66 A5   ¿f`â∞,ë╓ë╥╣..≤fÑ
00000B00   66 61 0F A9 0F A1 07 1F 66 9D CD 00 66 9C 1E 06   fa.⌐.í..f¥═.f£..
00000B10   0F A0 0F A8 66 60 FC 66 0F B7 E4 8C C8 8E D8 8E   .á.¿f`ⁿf.╗Σî╚Ä╪Ä
00000B20   C0 67 8B 7C 24 44 21 FF 74 08 89 E6 B9 0B 00 F3   └gï│$D! t.ëµ╣..≤
00000B30   66 A5 83 C4 2C 66 61 0F A9 0F A1 66 9D 66 C3 66   fÑâ─,fa.⌐.íf¥f├f
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C9200_9300_9400_95... × C9200_9300_9400_95... ×

```
00000B40   56 66 53 66 83 EC 2C 66 89 C3 66 83 F8 0A 75 0C   VfSfâ∞,ë├fâ°.u.
00000B50   66 B8 0D 00 00 00 66 E8 E3 FF FF FF 66 89 E0 66   f╕....fΦπ   fëαf
00000B60   E8 11 15 00 00 67 C7 44 24 10 07 00 67 C7 44 24   Φ....g╢D$...g╢D$
00000B70   18 01 00 67 C6 44 24 1D 0E 67 88 5C 24 1C 66 31   ...g╞D$..gê\$.f1
00000B80   C9 66 89 E2 66 B8 10 00 00 00 66 E8 50 FF FF FF   ╔fë┬f╕....fΦP
00000B90   66 8B 16 60 5F 66 85 D2 74 22 66 B9 FF FF 00 00   fï.`_fà╥t"f╣  ..
00000BA0   66 89 D6 66 83 C2 05 EC A8 20 74 08 88 D8 66 89   fë╓fâ┬.∞¿ t.ê╪fë
00000BB0   F2 EE EB 08 66 49 74 F4 F3 90 EB EB 66 83 C4 2C   ≥εδ.fIt⌡≤Éδδfâ─,
00000BC0   66 5B 66 5E 66 C3 66 53 66 89 C3 67 66 0F BE 03   f[f^f├fSfë├gf.╝.
00000BD0   84 C0 74 0A 66 43 66 E8 63 FF FF FF EB ED 66 5B   ä└t.fCfΦc   δφf[
00000BE0   66 C3 07 4E 6F 20 73 65 74 75 70 20 73 69 67 6E   f├.No setup sign
00000BF0   61 74 75 72 65 20 66 6F 75 6E 64 2E 2E 2E 0A 00   ature found.....
00000C00   66 BA A1 86 01 00 66 B9 20 00 00 00 66 4A 74 1F   f│íå..f╣ ...fJt.
00000C10   E6 80 E4 64 3C FF 75 04 66 49 74 13 A8 01 74 06   µÇΣd< u.fIt.¿.t.
00000C20   E6 80 E4 60 EB E6 A8 02 75 E2 66 31 C0 66 C3 66   µÇΣ`δµ¿.uΓf1└f├f
00000C30   83 C8 FF 66 C3 66 53 66 31 D2 8E E2 66 83 CA FF   â╙ f├fSf1╥ΓΓfâ╩
00000C40   8E EA 64 66 8B 0E 00 02 66 89 CA 67 66 8D 1C 08   ÄΩdfï...fë╩gfì..
00000C50   66 39 DA 74 16 66 42 64 66 89 16 00 02 E6 80 65   f9┌t.fBdfë...µÇe
00000C60   66 A1 10 02 66 31 D0 74 E7 EB 03 66 31 C0 64 66   fí..f1╨tτδ.f1└df
00000C70   89 0E 00 02 66 5B 66 C3 66 56 66 53 66 83 EC 2C   ë...f[f├fVfSfâ∞,
00000C80   66 BB 00 01 00 00 66 4B 0F 84 C2 00 66 B8 20 00   f╗....fK.ä┬.f╕ .
00000C90   00 00 66 E8 9D FF FF FF 66 85 C0 74 06 66 31 C0   ..fΦ¥   fà└t.f1└
00000CA0   E9 AF 00 66 89 E0 66 E8 CA 13 00 00 67 C7 44 24   Θ».fëαfΦ╩...g╢D$
00000CB0   1C 01 24 66 31 C9 66 89 E2 66 B8 15 00 00 00 66   ..$f1╔fë┬f╕....f
00000CC0   E8 1B FE FF FF 66 B8 20 00 00 00 66 E8 64 FF FF   Φ.▪  f╕ ...fΦd
00000CD0   FF 66 85 C0 75 C7 66 E8 24 FF FF FF 66 89 C6 66    fà└u╢├fΦ$   fë├f
00000CE0   B8 20 00 00 00 66 E8 4A FF FF FF 66 85 C0 75 AD   ╕ ...fΦJ   fà└u¡
00000CF0   66 85 F6 74 21 E4 92 66 83 E0 FE 66 83 C8 02 E6   fà÷t!ΣÆfâα▪fâ╙.µ
00000D00   92 66 B8 00 00 20 00 66 E8 28 FF FF FF 66 85 C0   Æf╕.. .fΦ(   fà└
00000D10   0F 84 72 FF EB 87 66 E8 E4 FE FF FF B0 D1 E6 64   .är δçfΦΣ▪  ╝╦µd
00000D20   66 E8 DA FE FF FF B0 DF E6 60 66 E8 D0 FE FF FF   fΦ┌▪  ╝■µ`fΦ╨▪
00000D30   B0 FF E6 64 66 E8 C6 FE FF FF 66 B8 00 00 20 00   ╝ µdfΦ╞▪  f╕.. .
00000D40   66 E8 EF FE FF FF 66 85 C0 74 AA E9 4F FF 66 83   fΦ∩▪  fà└t¬ΘO fâ
00000D50   C8 FF 66 83 C4 2C 66 5B 66 5E 66 C3 66 55 66 57   ╙ fâ─,f[f^f├fUfW
00000D60   66 56 66 53 66 83 EC 0C 67 66 89 54 24 04 66 85   fVfSfâ∞.gfëT$.fà
00000D70   C0 0F 84 D6 00 66 89 C2 66 83 E2 0F 66 C1 E8 04   └.ä╓.fë┬fâΓ.f┴Φ.
```

```
C9200_9300_9400_95...   C9200_9300_9400_95...
00000D80   8E E0 67 66 89 14 24 66 89 CF 66 31 D2 66 31 ED   Äαgfë.$fë╧f1╥f1φ
00000D90   66 83 C8 FF 67 66 8B 74 24 20 67 66 8D 5E FF 67   fâ╙ gfït$ gfì^ g
00000DA0   66 89 5C 24 08 67 66 8B 1C 24 64 67 8A 1B 84 DB   fë\$.gfï.$dgè.ä█
00000DB0   0F 84 88 00 66 83 FA 02 74 3F 66 83 FA 03 74 40   .äê.fâ·.t?fâ·.t@
00000DC0   66 4A 74 0B 66 31 D2 80 FB 20 76 5E E9 82 00 66   fJt.f1╥Ç√ v^Θé.f
00000DD0   89 EE 80 FB 3D 75 08 67 80 3E 00 74 39 EB 08 66   ëεÇ√=u.gÇ>.t9δ.f
00000DE0   31 D2 80 FB 20 76 43 67 66 8D 6E 01 66 31 D2 67   1╥Ç√ vCgfìn.f1╥g
00000DF0   3A 1E 0F 95 C2 66 42 EB 31 80 FB 20 77 2C EB 27   :..ò┬fBδ1Ç√ w,δ'
00000E00   80 FB 20 76 22 67 66 39 44 24 08 7E 05 67 88 1F   Ç√ v"gf9D$.~.gê.
00000E10   66 47 66 40 EB 14 66 89 CF 66 89 F5 66 BA 03 00   fGf@δ.fë╧fë│f╢..
00000E20   00 00 66 31 C0 EB 03 66 31 D2 67 66 FF 04 24 67   ..f1└δ.f1╥gf .$g
00000E30   66 81 3C 24 00 00 01 00 0F 85 69 FF 67 66 83 7C   fü<$.....àï gfâ|
00000E40   24 20 00 74 1A 67 C6 07 00 EB 14 66 83 C8 FF EB   $ .t.g╞..δ.fâ╙ δ
00000E50   0E 67 66 8B 74 24 04 80 FB 3D 75 8B E9 78 FF 66   .gfït$.Ç√=uïΘx f
00000E60   83 C4 0C 66 5B 66 5E 66 5F 66 5D 66 C3 66 55 66   â─.f[f^f_f]f├fUf
00000E70   57 66 56 66 53 66 51 67 66 89 14 24 66 85 C0 0F   WfVfSfQgfë.$fà└.
00000E80   84 86 00 66 89 C7 66 83 E7 0F 66 C1 E8 04 8E E0   äå.fë╟fâτ.f┴Φ.Äα
00000E90   66 89 F9 66 31 D2 66 31 DB 66 31 C0 67 66 8D 71   fë·f1╥f1╦f1└gfìq
00000EA0   01 64 67 8A 09 66 89 F5 66 29 FD 66 83 FA 01 74   .dgè.fë│f)²fâ·.t
00000EB0   17 66 83 FA 02 74 38 84 C9 74 54 80 F9 20 76 38   .fâ·.t8ä╔tTÇ· v8
00000EC0   67 66 8B 1C 24 66 89 E8 67 8A 13 84 D2 75 0D 80   gfï.$fëΦgè.ä╥u.Ç
00000ED0   F9 20 76 3E 66 BA 02 00 00 00 EB 1F 84 C9 74 2F   · v>f╢....δ.ä╔t/
00000EE0   66 43 38 D1 0F 95 C2 66 0F B6 D2 66 42 EB 0C 84   fC8╤.ò┬f.╢╥fBδ.ä
00000EF0   C9 74 1C 80 F9 20 77 03 66 31 D2 66 89 F1 66 81   ╔t.Ç· w.f1╥fë±fü
00000F00   FE 00 00 01 00 75 95 EB 06 66 83 C8 FF EB 03 66   ▪....uòδ.fâ╙ δ.f
00000F10   31 C0 66 5A 66 5B 66 5E 66 5F 66 5D 66 C3 56 57   1└fZf[f^f_f]f├VW
00000F20   89 C7 89 D6 51 C1 E9 02 F3 66 A5 59 83 E1 03 F3   ë╟ë╓Q┴Θ.≤fÑYâß.≤
00000F30   A4 5F 5E 66 C3 57 89 C7 66 0F B6 C2 66 69 C0 01   ñ_^f├Wë╟f.╢┬fi└.
00000F40   01 01 01 51 C1 E9 02 F3 66 AB 59 83 E1 03 F3 AA   ...Q┴Θ.≤f½Yâß.≤¬
00000F50   5F 66 C3 1E 0F A0 1F 66 E8 C1 FF FF FF 1F 66 C3   _f├..á.fΦ┴   .f├
00000F60   06 0F A0 07 66 E8 B4 FF FF FF 07 66 C3 66 55 66   ..á.fΦ┤   .f├fUf
00000F70   57 66 56 66 53 66 83 EC 10 67 66 8D 4C 24 04 67   WfVfSfâ∞.gfìL$.g
00000F80   66 8D 54 24 0C 67 66 8D 44 24 08 66 E8 F8 02 00   fìT$.gfìD$.fΦ°..
00000F90   00 67 66 8B 44 24 0C 67 66 39 44 24 08 0F 8D 99   .gfïD$.gf9D$..ìÖ
00000FA0   00 66 BA E0 2D 00 00 66 83 F8 40 74 2A 66 83 F8   .f╢α-..fâ°@t*fâ°
00000FB0   0F 75 06 66 B8 06 00 00 00 66 50 66 68 E7 2D 00   .u.f╕....fPfhτ-.
```

```
00000FC0   00 66 68 60 3C 00 00 66 E8 51 10 00 00 66 83 C4   .fh`<..fΦQ...fâ─
00000FD0   0C 66 BA 60 3C 00 00 66 52 66 68 ED 2D 00 00 66   .f╢`<..fRfhφ-..f
00000FE0   E8 53 10 00 00 67 66 8B 44 24 10 66 5E 66 5F 66   ΦS...gfïD$.f^f_f
00000FF0   BA E0 2D 00 00 66 83 F8 40 74 2A 66 83 F8 0F 75   ╢α-..fâ°@t*fâ°.u
00001000   06 66 B8 06 00 00 00 66 50 66 68 E7 2D 00 00 66   .f╕....fPfhτ-..f
00001010   68 60 3C 00 00 66 E8 03 10 00 00 66 83 C4 0C 66   h`<..fΦ....fâ─.f
```

During manual inspection of the beginning of the firmware file in a hex editor, several readable metadata strings were observed.

In the first part of the file there are multiple entries starting with KEY_TLV, for example:

KEY_TLV_PACKAGE_COMPATIBILITY

KEY_TLV_PACKAGE_BOOTARCH

KEY_TLV_BOARD_COMPAT

KEY_TLV_CRYPTO_KEYSTRING

These entries likely describe compatibility information for the firmware such as supported hardware platform, board type and boot architecture.

Further in the header there are additional metadata fields describing the firmware itself:

CW_BEGIN=$$
CW_FAMILY=$cat9k$
CW_IMAGE=$cat9k_iosxe.16.12.07.SPA.bin$
CW_VERSION=$16.12.07$
CW_FULL_VERSION=$16.12.07.0.6565.1643829996$

CW_DESCRIPTION=$Cisco IOS Software, IOS-XE Software$
CW_END=$$

These fields appear to contain general information about the firmware image, including its name, version and description.

Right after this section there are strings that look like certificate information, for example:

CN=CiscoSystems
OU=IOS-XE
O=CiscoSystems

This likely indicates that the firmware image contains a signature or certificate block used for verification.

After this metadata area the file transitions into mostly binary data, which likely corresponds to executable code or compressed firmware components.